SWALR#

https://pytorch.org/docs/stable/generated/torch.optim.swa_utils.SWALR.html#t

Description:

Anneals the learning rate in each parameter group to a fixed value. This learning rate scheduler is meant to be used with Stochastic Weight Averaging (SWA) method (see torch.optim.swa_utils.AveragedModel). optimizer (torch.optim.Optimizer) – wrapped optimizer swa_lrs (float or list) – the learning rate value for all param groups together or separately for each group. annealing_epochs (int) – number of epochs in the annealing phase (default: 10)

Function Signature

```
class torch.optim.swa_utils.SWALR(optimizer, swa_lr,
anneal_epochs=10, anneal_strategy='cos', last_epoch=-1)
[source]#

Parameters

get_last_lr()[source]#

Return type

get_lr()[source]#
```

Returns:

dict[str, Any]

Code Examples

```
>>> loader, optimizer, model = ...
>>> lr_lambda = lambda epoch: 0.9
>>> scheduler =
torch.optim.lr_scheduler.MultiplicativeLR(optimizer,
>>>         lr_lambda=lr_lambda)
>>> swa_scheduler = torch.optim.swa_utils.SWALR(optimizer,
>>>         anneal_strategy="linear", anneal_epochs=20,
swa_lr=0.05)
>>> swa_start = 160
>>> for i in range(300):
>>>      for input, target in loader:
>>>          optimizer.zero_grad()
>>>          loss_fn(model(input), target).backward()
>>>          optimizer.step()
>>>      if i > swa_start:
>>>          swa_scheduler.step()
>>>      else:
>>>          scheduler.step()
```

Detailed Documentation

Documentation

Anneals the learning rate in each parameter group to a fixed value.

This learning rate scheduler is meant to be used with Stochastic Weight Averaging (SWA) method (see torch.optim.swa_utils.AveragedModel).

optimizer (torch.optim.Optimizer) – wrapped optimizer

swa_lrs (float or list) – the learning rate value for all param groups together or separately for each group.

annealing_epochs (int) – number of epochs in the annealing phase (default: 10)

annealing_strategy (str) – "cos" or "linear"; specifies the annealing strategy: "cos" for cosine annealing, "linear" for linear annealing (default: "cos")

last_epoch (int) – the index of the last epoch (default: -1)

The SWALR scheduler can be used together with other schedulers to switch to a constant learning rate late in the training as in the example below.

Return last computed learning rate by current scheduler.

Load the scheduler's state.

state_dict (dict) – scheduler state. Should be an object returned from a call to state_dict().

Return the state of the scheduler as a dict.

It contains an entry for every variable in self.__dict__ which is not the optimizer.